

### CPU Scheduling Problem (FCFS)

Consider the following set of processes:

| Process | Arrival Time (AT) | Burst Time (BT) |
|---------|-------------------|-----------------|
| P1      | 0                 | 5               |
| P2      | 2                 | 3               |
| P3      | 4                 | 6               |
| P4      | 6                 | 4               |
| P5      | 8                 | 2               |

Using the **First Come First Serve (FCFS)** scheduling algorithm, calculate the following:

1. **Gantt Chart**
2. For each process, compute:
  - Completion Time (CT)
  - Turnaround Time (TAT) = CT – AT
  - Waiting Time (WT) = TAT – BT
  - Response Time (RT)
3. Compute the following averages:
  - Average Turnaround Time
  - Average Waiting Time
  - Average Response Time
4. Also calculate:
  - **Throughput** (Total number of processes / Total time)
  - **Scheduling Length** (Total time taken to execute all processes)

### CPU Scheduling Problem (SJF - Non-Preemptive)

Consider the following processes:

| Process | Arrival Time (AT) | Burst Time (BT) |
|---------|-------------------|-----------------|
| P1      | 0                 | 7               |
| P2      | 2                 | 4               |
| P3      | 4                 | 1               |
| P4      | 5                 | 4               |
| P5      | 6                 | 2               |

Using the **Shortest Job First (SJF - Non-Preemptive)** scheduling algorithm, perform the following:

1. Draw the **Gantt Chart**
2. For each process, calculate:
  - Completion Time (CT)
  - Turnaround Time (TAT) =  $CT - AT$
  - Waiting Time (WT) =  $TAT - BT$
  - Response Time (RT)
3. Compute:
  - Average Turnaround Time
  - Average Waiting Time
  - Average Response Time
4. Also determine:
  - **Throughput** = Total number of processes / Total execution time
  - **Scheduling Length** = Time from first process start to last process completion